

Machine Learning Unit 2 Notes: Supervised Learning

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Subject: Machine Learning | Unit 2: Supervised Learning Algorithms

1. Linear Regression and Cost Functions:

Linear regression models the relationship between dependent continuous variables and independent explanatory features using the ordinary least squares or gradient descent optimization.

2. Classification and Logistic Regression:

Logistic regression applies the sigmoid activation function to output predicted class probabilities between 0 and 1.

3. Decision Trees and Random Forests:

Decision trees partition the feature space based on Information Gain or Gini Impurity. Ensemble learning combines multiple weak learners to decrease variance and prevent overfitting.